about electronics can make easy and simple learning process. The purpose of this chapter is to give basic elementary knowledge in order to understand the following chapters.

What is meant by electronics..?

The branch of engineering which deals with current conduction through a vacuum or gas or semiconductor is known as electronics.

Few important activities of electronic devices are

1. Rectification
2. Amplification
3. Control
4. Oscillations
5. Conversion of light into electricity and
6. Conversion of electricity into light etc.

The first step to understand the principles of electronics starts from knowing about an atom, since everything is made up of atom.

3.1. ATOMIC STRUCTURE

According to modern theory, matter is electrical in nature. All the materials are composed of very tiny particles called atoms. The atoms are the root cause for all the matter or material existing in this world.

The atom consists of a central nucleus, contains protons and neutrons as shown in Figure 3.1. A proton is a positively charged particle, where neutron does not have any charge. There is yet another particle called electrons which is negative in charge and it is not reside inside the nucleus rather revolving around the nucleus. This is termed as extra nucleus.

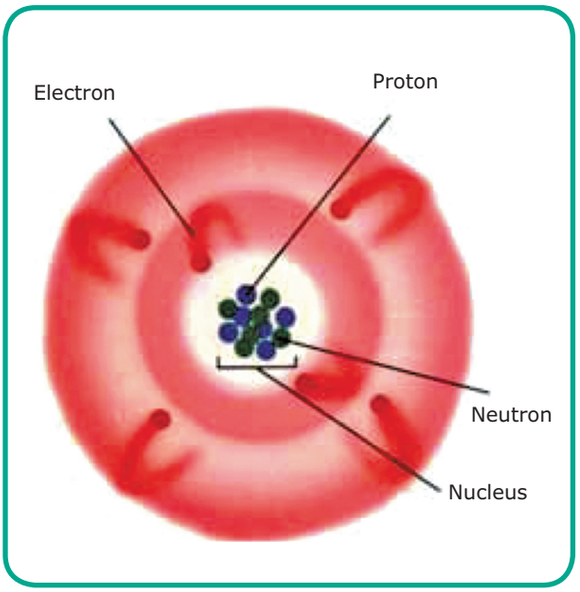


Figure 3.1 Atomic structure

The character of any atoms can be defined by three factors.

1. Atomic number
2. Atomic weight
3. Electrical charge

3.1.1 Nucleus

It is a central part of an atom, which contains protons and neutrons. A proton is a positively charged particle, while the neutron has the same 'mass' as the proton , but has no charge. Therefore the nucleus of an atom is positively charged. The sum of protons and neutrons decides the entire weight of an atom. Because, the particle in the extra nucleus (i.e., electrons) have negligible weight as compared to protons and neutrons.

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